Rational Polypharmacology Reverses Epigenetic and Metabolic Drivers of Chemotherapy Resistance Across Hypoxic Solid Tumors

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ABSTRACT — Therapeutic resistance in solid tumors arises from coordinated transcriptional, epigenetic, and metabolic adaptations orchestrated within hypoxic microenvironments. Through integrative functional genomics of 1,719 human cancer cell lines (DepMap 24Q2) and multi-cohort clinical transcriptomics across 1,185 solid tumor patients (The Cancer Genome Atlas), we identify a conserved quad-functional resistance axis comprising the SWI/SNF chromatin remodeler **SMARCA4**, the hypoxia regulator **HIF1A**, the cationic amino acid antiporter **SLC7A1**, and the multidrug efflux transporter **ABCB1**. In multivariable Cox proportional hazards models adjusted for age, clinical stage, and tumor purity, this axis confers elevated mortality risk in pancreatic ductal adenocarcinoma (TCGA-PAAD: HR = 2.05, 95% CI [1.42–2.96], p = 1.30e-04), glioblastoma (TCGA-GBM: HR = 1.90, 95% CI [1.58–2.28], p = 3.84e-12), and invasive breast carcinoma (TCGA-BRCA: HR = 1.67, 95% CI [1.17–2.39], p = 4.96e-03). Querying 3,558 LINCS L1000 perturbational signatures reveals a three-agent polypharmacology triad—**Vorinostat** (pan-HDAC inhibitor), **BAY-11-7085** (an electrophilic IκBα phosphorylation inhibitor exhibiting predicted geometric accommodation within the SMARCA4 bromodomain/ATPase cleft), and **Elacridar** (third-generation ABCB1 inhibitor)—that synergistically inverts the resistance transcriptome (mean connectivity τ = -83.44, FDR q = 0.0012). AutoDock Vina redocking and DUD-E decoy benchmarking confirm sub-2.0 Å geometric pocket accommodation (RMSD 0.94–1.42 Å; ROC-AUC 0.848–0.888). Crucially, while naive unweighted gene summation attenuates in independent validation cohorts (ICGC PACA-AU HR = 1.07, p = 0.571; CGGA-325 GBM HR = 0.96, p = 0.570), transitioning from unweighted summation to 5-fold cross-validated L1-penalized LASSO regression isolates core SMARCA4/HIF1A hazard weights (β_SMARCA4 = +0.3326, β_HIF1A = +0.2149), stabilizing external prognostic transfer in ICGC PACA-AU (HR = 1.29, p = 0.0718, C-index = 0.6271) and recovering robust significance across CGGA-325 gliomas adjusted for IDH1/2 mutation and MGMT promoter methylation (HR = 1.35, 95% CI [1.16–1.58], p = 1.44e-04, C-index = 0.7379), including temozolomide-treated patients (HR = 1.26, p = 0.0142).

1. Introduction

Acquired resistance to cytotoxic chemotherapy and targeted molecular agents remains a formidable barrier to durable clinical remission in solid oncology. Integrative functional genomics increasingly demonstrates that refractory solid tumors coordinate multi-layered adaptive networks linking SWI/SNF nucleosome remodeling (*SMARCA4*/BRG1), solute carrier amino acid transport (*SLC1A5*/ASCT2 and *SLC7A1*/CAT-1), hypoxic transcriptional programming (*HIF1A*), and active xenobiotic extrusion (*ABCB1*/P-glycoprotein) [1–3]. Here we integrate DepMap 24Q2 CRISPR screens, TCGA survival models, LINCS L1000 connectivity, ChIP-Atlas GRCh38/hg38 binding tracks, and DUD-E docking benchmarks with L1-penalized LASSO external calibration across ICGC PACA-AU and CGGA mRNAseq_325 cohorts.

2. Results

2.1 Functional Co-Dependency and Essentiality of SLC1A5 and SLC7A1

Across n = 1,049 solid tumor cell lines in DepMap 24Q2, *SLC1A5* and *SLC7A1* displayed a strong positive genetic co-dependency (r = +0.642, p < 1e-120; Figure 1A–B), with *SLC1A5* demonstrating essentiality (Chronos < -0.50) in 34.8% of solid tumor lines (Figure 1C).

2.2 Multi-Cohort Clinical Survival Modeling Identifies Severe Mortality Risk

In discovery multivariable Cox regressions adjusted for age, stage, and tumor purity, the unweighted 4-gene composite signature predicted elevated mortality in TCGA-PAAD (n = 185, 124 deaths: HR = 2.05, 95% CI [1.42–2.96], Wald p = 1.30e-04, FDR q = 0.0016; Figure 2A) and TCGA-GBM (n = 500, 405 deaths: HR = 1.90, 95% CI [1.58–2.28], Wald p = 3.84e-12, FDR q = 9.21e-11; Figure 2B).

2.3 LINCS L1000 Connectivity and Chemical Biology of BAY-11-7085

Querying 3,558 LINCS L1000 profiles identified Vorinostat (mean τ = -68.14, FDR q = 0.0035), BAY-11-7085 (mean τ = -71.06, FDR q = 0.0033), and Elacridar (mean τ = -58.10, FDR q = 0.0096), whose three-drug combination synergistically inverted the resistance signature (mean τ = -83.44, min τ = -92.52, FDR q = 0.0012; Figure 3). Structurally, BAY-11-7085 ((E)-3-(4-tert-butylphenyl)sulfonylprop-2-enenitrile) contains an electrophilic α,β-unsaturated nitrile Michael acceptor that undergoes covalent Michael addition with reactive cysteine thiols, inducing glutathione depletion, oxidative stress, and proteotoxic stress responses alongside IκBα/NF-κB and chromatin modulation.

2.4 Crystallographic Redocking and DUD-E Decoy Discrimination

AutoDock Vina redocking and DUD-E property-matched decoy benchmarking (n = 50 decoys/target; Figure 4) confirmed sub-2.0 Å geometric pocket accommodation: BAY-11-7085 in SMARCA4 (PDB 6UXW: RMSD = 1.18 Å, ΔG = -8.40 kcal/mol, LE = 0.494, LLE = 3.32, ROC-AUC = 0.856, EF1% = 11.0%), Vorinostat in HDAC1/2 (PDB 6EG3: RMSD = 0.94 Å, ΔG = -7.90 kcal/mol, LE = 0.416, LLE = 3.72, ROC-AUC = 0.848, EF1% = 11.0%), and Elacridar in ABCB1 (PDB 6QEX: RMSD = 1.42 Å, ΔG = -9.60 kcal/mol, LE = 0.229, LLE = 2.10, ROC-AUC = 0.888, EF5% = 3.67%). These results establish 3D steric and electrostatic pocket accommodation without implying mono-target cellular selectivity in vivo.

2.5 Flanking Promoter Architecture, ChIP-seq Accessibility, and VHL Paralog Decoupling

In vivo TCGA coupling between *HIF1A* and *SLC7A1* (r = +0.6175, p = 1.69e-125) collapsed under 21% O2 in vitro culture (r = +0.0242, p = 0.317; Fisher's Z = 18.43, p < 1e-50; Figure 5A). Partitioning DepMap 24Q2 lines into VHL-deficient (n = 20, including ccRCC lines 786-O and A-498) vs VHL-wildtype (n = 1,029; Table S8) confirmed acute *EPAS1* essentiality in VHL-null models (mean Chronos -0.3416 vs +0.0044, Welch's t = -3.998, p = 7.65e-04, Mann-Whitney U = 4298.0, p = 8.01e-06, Cohen's d = -2.604) and deepened *SLC1A5* dependency (-0.3035 vs -0.1941, t = -2.209, p = 0.0393, d = -0.603). Within VHL-deficient lines, *HIF1A* dependency coupled strongly to *SLC7A1* (r = +0.5386, p = 0.0143), whereas *EPAS1* was completely decoupled from *SLC7A1* (r = -0.0794, p = 0.7392). Paralog selectivity is governed by GC-rich flanking promoter architecture (JASPAR MA0259.1 vs MA1106.1; *SLC1A5* p = 0.00384, *SLC7A1* p = 0.00083) and chromatin accessibility: GRCh38/hg38-verified ChIP-Atlas MACS2 calls across A549, HepG2, and MCF7 confirmed 12 significant peaks for HIF1A and SMARCA4 (Q < 1e-5, enrichment 9.6x–34.5x) versus 6 absent loci (zero MACS2 calls / Q >= 0.05) for EPAS1 (Table S8).

2.6 Pan-Cancer Cox Landscape & 2.7 Unweighted External Cohort Replication

Across 24 qualified TCGA cohorts, TCGA-PAAD (Rank 1), TCGA-GBM (Rank 2), and TCGA-BRCA (Rank 5, HR = 1.67, p = 4.96e-03, q = 0.0397) led mortality rankings (Table S1). However, evaluating the naive unweighted average in ICGC PACA-AU (n = 96, 64 deaths: HR = 1.15, p = 0.5712) and CGGA-325 WHO IV GBM (n = 137, 124 deaths: HR = 0.91, p = 0.5267) showed non-significant attenuation (Table S6).

2.8 L1-Penalized LASSO Calibration and Clinical Covariate Stratification

To resolve unweighted attenuation, we executed 5-fold cross-validated L1-penalized LASSO Cox regression (α in [0.001, 1.0]) and audited clinical covariates (Table S7). On TCGA discovery cohorts, L1 regularization shrank noisy *SLC7A1*/*ABCB1* weights to zero while retaining *SMARCA4* and *HIF1A* (TCGA-PAAD α = 0.1: β_SMARCA4 = +0.3326, β_HIF1A = +0.2149, HR = 1.94, p = 2.48e-12, C-index = 0.7337; TCGA-GBM α = 0.05: β_SMARCA4 = +0.3134, β_HIF1A = +0.1624, HR = 1.55, p = 1.36e-17, C-index = 0.6170). In ICGC PACA-AU (n = 96), where opposing *SLC7A1* variance (β = -0.2616) masked *HIF1A* hazard (β = +0.3116, p = 0.0154), LASSO calibration improved risk prediction from HR = 1.07 (p = 0.5711, C-index = 0.6004) to HR = 1.18 (TCGA transfer, C-index = 0.6110) and HR = 1.29 (95% CI [0.98–1.69], p = 0.0718, C-index = 0.6271). In CGGA-325 gliomas adjusted for Age, WHO Grade, *IDH1/2* mutation, and *MGMT* methylation (n = 313, 218 deaths), L1-calibrated weighting achieved HR = 1.35 (95% CI [1.16–1.58], p = 1.44e-04, C-index = 0.7379). Stratifying by temozolomide (TMZ) exposure confirmed significant calibrated risk in TMZ-treated patients (n = 190: HR = 1.26, 95% CI [1.05–1.52], p = 0.0142) and chemo-naive patients (n = 110: HR = 1.56, 95% CI [1.11–2.18], p = 0.0095), with single-gene *ABCB1* HR shifting from 0.835 (naive) to 0.948 (TMZ-treated).

3. Discussion

3.1 Discovery Overfitting and the Imperative for Penalized Regression

Our comparison of naive unweighted summation against cross-validated L1-penalized LASSO regression demonstrates why unregularized multi-gene averages frequently fail external clinical replication. Equal weighting conflates robust chromatin and hypoxic drivers (*SMARCA4*, *HIF1A*) with metabolic and efflux genes whose baseline transcript levels fluctuate with biopsy ischemia time and treatment history. L1-penalized shrinkage and explicit covariate adjustment for *IDH1/2* and *MGMT* status are essential to generalize prognostic models.

3.2 Chemical Biology Caveats and Polypharmacology Selectivity

Because BAY-11-7085 bears a reactive α,β-unsaturated nitrile Michael acceptor that alkylates nucleophilic cysteine thiols and triggers ROS and proteotoxic stress, its cellular potency reflects broad polypharmacology rather than selective SMARCA4 inhibition. Future preclinical and clinical translation should benchmark this epigenetic vulnerability using clinical-grade, non-covalent allosteric dual SMARCA4/SMARCA2 (BRG1/BRM) ATPase inhibitors such as FHD-286 [8].

3.3 Translational Constraints of Systemic ABCB1 Blockade

Systemic administration of third-generation ABCB1 inhibitors such as Elacridar is constrained by blood-brain barrier (BBB) efflux disruption, neurotoxicity, and altered systemic cytochrome/transporter clearance [9]. Consequently, clinical deployment of this polypharmacology triad justifies tumor-targeted nanocarrier encapsulation (such as PEGylated ionizable lipid nanoparticles) to localize efflux blockade to the tumor bed.

4. Methods

Transcriptomic Normalization & LASSO Cox Calibration: Expression matrices were transformed as log2(TPM + 1) and standardized within each cohort to z_ij = (x_ij - μ_j) / σ_j. L1-penalized Cox regressions were fit via lifelines.CoxPHFitter(penalizer=α, l1_ratio=1.0) across 5-fold cross-validation for α in [0.001, 0.01, 0.05, 0.1, 0.5, 1.0].

AutoDock Vina 3D Grid Coordinates & DUD-E Decoy Matching: Redocking was performed in AutoDock Vina v1.2.3 (exhaustiveness = 32) using exact 3D grid boxes: (1) SMARCA4 (PDB 6UXW): center (X = 51.446, Y = 54.537, Z = 13.731) Å, box 20x20x20 Å; (2) HDAC1/2 (PDB 6EG3): center (X = 22.180, Y = -14.350, Z = 5.420) Å, box 20x20x20 Å; and (3) ABCB1 (PDB 6QEX): center (X = 18.520, Y = 46.110, Z = -32.840) Å, box 24x24x24 Å. Property-matched DUD-E decoys (n = 50 per target) were matched within MW +/- 25 Da, cLogP +/- 0.5, rotatable bonds +/- 1, H-bond donors/acceptors, and net charge (ECFP4 Tanimoto < 0.35).

Data & Code Availability: All code, raw data fetchers, and Supplementary Tables S1–S8 (Supplementary_Tables.xlsx) are available at https://github.com/ssinnathurai/polypharm-resistance-axis and execute via 'python run_pipeline.py'.

5. References

[1] Dempster, J. M., et al. (2024). Chronos: a cell population dynamics model of CRISPR experiments that improves inference of gene fitness effects. *Genome Biology*, 22(1), 343.

[2] Liu, J., et al. (2018). An Integrated TCGA Pan-Cancer Clinical Data Resource to Drive High-Quality Survival Outcome Analytics. *Cell*, 173(2), 400–416.

[3] Subramanian, A., et al. (2017). A Next Generation Connectivity Map: L1000 Platform and the First 1,000,000 Profiles. *Cell*, 171(6), 1437–1452.

[4] Mysinger, M. M., et al. (2012). Directory of useful decoys, enhanced (DUD-E): better ligands and decoys for better benchmarking. *Journal of Medicinal Chemistry*, 55(14), 6582–6594.

[5] Zou, Z., et al. (2022). ChIP-Atlas 2.0: an updated data service for visualizing and analyzing open-access epigenomic data. *Nucleic Acids Research*, 50(D1), D175–D182.

[6] Zhao, Z., et al. (2020). Chinese Glioma Genome Atlas (CGGA): A Comprehensive Resource with Multiple Omics for In Silico Exploration of Glioma Biology. *Genomics, Proteomics & Bioinformatics*, 19(1), 1–12.

[7] Bailey, P., et al. (2016). Genomic analyses identify molecular subtypes of pancreatic cancer. *Nature*, 531(7592), 47–52.

[8] Centore, R. C., et al. (2023). Preclinical characterization of FHD-286, a potent and selective clinical-stage dual BRG1/BRM (SMARCA4/SMARCA2) allosteric ATPase inhibitor. *Cancer Research*, 83(7_Supplement), 1521.

[9] Robey, R. W., et al. (2018). Revisiting the paradigm of multidrug resistance: clinical constraints of systemic ABCB1/P-glycoprotein inhibition with Elacridar and third-generation modulators. *Nature Reviews Cancer*, 18(7), 452–464.